LESSON PLAN

AIRCRAFT HUSBANDRY, CORROSION PREVENTION AND SURFACE PROTECTION (25)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	02
Total No. of Lesson Plans	7	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Identification of AN-type, NAS etc.; grip length, torques, a/c bolt identification, screws, taper pins.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Identification of AN-type, NAS etc.; grip length, torques, a/c bolt identification, screws, taper pins.
5. Specific Objectives	25.1.1.1 Trainee will be able to describe the use of various standards, identification of bolts, nuts, safety methods and maintenance of cables.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Identification of AN-type, NAS, screws, and taper pins using physical samples; Safety precautions for handling fasteners; Instructor supervision during identification exercises.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the two primary standards for aircraft fasteners?
(b) Why is grip length critical in aircraft structural assembly?
(c) What is the primary function of a taper pin in aircraft maintenance?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• AN-type and NAS fastener identification • Grip length measurement techniques • Standard torque values for bolts	L & Ds	PPT, IB	15 min
(b)	• Aircraft bolt identification procedures • Screw types and applications • Taper pin specifications and usage	L & Ds	PPT, PDF	20 min
(c)	• Safety methods for fastener handling • Proper storage and maintenance protocols • Inspection criteria for damaged fasteners	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Differentiate between AN-type and NAS fasteners based on head markings.
(b) Explain the relationship between grip length and thread engagement in aircraft structures.
(c) Describe the correct procedure for selecting and installing a taper pin.

12. Home Assignment and Follow Up	Prepare a brief report comparing the torque specifications for AN-type bolts versus NAS bolts for a specific aircraft application covered in the next lesson plan.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Nuts, miscellaneous nuts, washers, Safety methods for turn buckles and various types of turn buckles.	Refer to the master syllabus reference list	LP 03

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min